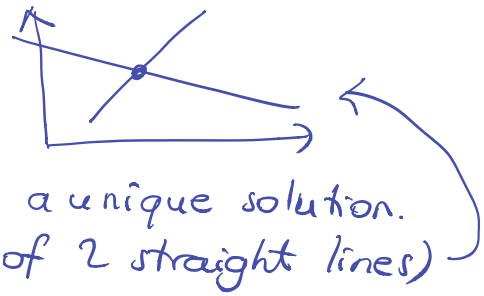
Ch13 - Eigenvalues

Non homogeneous Eq^{n}s:

$$\underline{\underline{A}}\,\underline{x} = \underline{b}$$

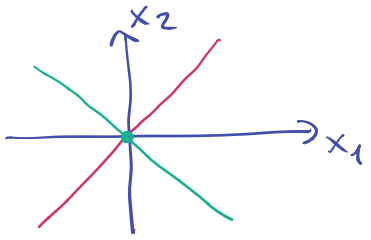


- If eq^{n}s are independent ⇒ a unique solution.
  (in 2D ⇒ intersection of 2 straight lines)

Homogeneous system:

$$\underline{\underline{A}}\,\underline{x} = \underline{0}$$

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 &= 0 \\ a_{21}x_1 + a_{22}x_2 &= 0 \end{aligned}\right\}$$



Two straight lines that intersect at zero.

Trivial solution : $\underline{x} = \underline{0}$

Eigenvalue Problems:

$$\underline{\underline{A}}\,\underline{x} = \lambda\,\underline{x}, \qquad \lambda : \text{eigenvalue}, \quad \underline{x} : \text{eigenvector}$$

$$\underline{\underline{A}}\,\underline{x} = \lambda\,\underline{\underline{I}}\,\underline{x} \qquad \underline{\underline{I}} : \text{identity matrix}$$

$$\underline{x}$$

$$\underline{\underline{A}}\,\underline{x} - \lambda\,\underline{\underline{I}}\,\underline{x} = \underline{0}$$

$$(\underline{\underline{A}} - \lambda\,\underline{\underline{I}})\,\underline{x} = \underline{0}$$

For nontrivial solution to exist, the determinant of the matrix must be zero:

$$\left|\underline{\underline{A}} - \lambda\,\underline{\underline{I}}\right| = 0$$

Expand the determinant $\Rightarrow$ get a polynomial in λ
(characteristic polynomial)

Two eq^n case:

$$\underline{\underline{A}} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$(a_{11}-\lambda)x_1 + x_2 = 0$$

$$a_{21}x_1 + (a_{22}-\lambda)x_2 = 0$$

$$A - \lambda I = \begin{bmatrix} a_{11}-\lambda & a_{12} \\ a_{21} & a_{22}-\lambda \end{bmatrix}$$

$$\begin{vmatrix} a_{11}-\lambda & a_{12} \\ a_{21} & a_{22}-\lambda \end{vmatrix} = 0 \Rightarrow (a_{11}-\lambda)(a_{22}-\lambda) - a_{12}a_{21} = 0$$

$$\lambda^2 - (a_{11}+a_{22})\lambda - a_{12}a_{21} = 0$$

$$\lambda_{1,2} = \frac{(a_{11}+a_{22}) \pm \sqrt{(a_{11}+a_{22})^2 - 4a_{12}a_{21}}}{2}$$

$\lambda_{1,2}$ ↙ 2 solutions

Ex

$$(10-\lambda)x_1 - 5x_2 = 0 \quad (1) \qquad 10x_1 - 5x_2 = 0$$

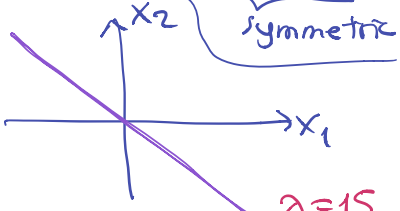
$$-5x_1 + (10-\lambda)x_2 = 0 \quad (2) \qquad -5x_2 + 10x_2 = 0$$

$$\begin{vmatrix} 10-\lambda & -5 \\ -5 & 10-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 20\lambda + 75 = 0$$

$\lambda_1 = 15$

$\lambda_2 = 5$

$A = \begin{bmatrix} 10 & -5 \\ -5 & 10 \end{bmatrix}$ symmetric



Put $\boxed{\lambda_1 = 15}$ into (1) and (2)

$$-5x_1 - 5x_2 = 0$$
$$-5x_1 - 5x_2 = 0.$$
} identical

infinite solutions: $x_2 = -x_1 \Rightarrow$ one solution: $x = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ ← eigenvector

$\lambda = 15$ Eq^{n}s 1 and 2

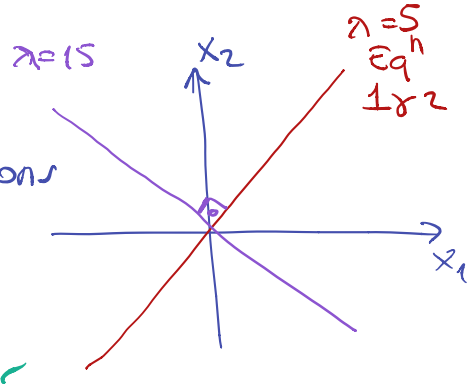
Put $\lambda_2 = 5$ into (1) and (2):

$$5x_1 - 5x_2 = 0$$
$$-5x_1 + 5x_2 = 0$$

> identical infinite solutions

$$x_2 = x_1$$

one solution : $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ → eigenvector



* Eigenvectors are orthogonal : a property of symmetric matrices with distinct eigenvalues.

* $Ax = \lambda x$
  information content of A is transformed into a scalar λ.